

Nguyen Vu Binh Duong

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EDUCATION

University of Engineering and Technology, Vietnam National University (VNU-UET) Hanoi, Vietnam

M.Sc. in Software Engineering 2023 – 2025

- **GPA:** 3.87/4.00
- **Master's Thesis:** AI-based Software System for Early Diagnosis and Detection of Alzheimer's Disease in Vietnamese People

University of Engineering and Technology, Vietnam National University Hanoi, Vietnam

B.Sc. in Computer Science 2019 – 2023

- **GPA:** 3.60/4.00
- **Bachelor's Thesis:** Improving the Performance of Unit Test Case Execution for C/C++ Projects

RESEARCH EXPERIENCE

International University, Vietnam National University Ho Chi Minh City HCMC, Vietnam

Brain Health Lab, School of Biomedical Engineering

Researcher 2024 – Present

- Alzheimer's Biomarker Cross-border Development (ABCD) Project, Vietnamese-Swiss Joint Research Programme.
Supervisor: Assoc. Prof. Huong Ha
 - * Developed a modular machine learning framework, comprising predictor and explainer modules, for Alzheimer's disease (AD) classification using structural MRI data from a Vietnamese cohort (N=113).
 - * Identified the Generalized Additive Model as the most interpretable and robust approach (78.8% accuracy, 87.2% AUC) for the Vietnamese cohort, balancing performance with clinical explainability.
 - * Conducted a comparative analysis of structural features between the Vietnamese cohort (N=76, from April 30th Hospital) and the ADNI cohort (N=80).
 - * Showed that cross-dataset validation resulted in an approximate 10% to 25% drop in prediction accuracy, highlighting the need for population-specific ML models.

University of Engineering and Technology, Vietnam National University Hanoi, Vietnam

Department of Software Engineering, Faculty of Information Technology

Lab Instructor/Research Assistant 2023 – Present

Undergraduate Research Assistant 2021 – 2023

- AKAUTAUTO: a proprietary software used for C/C++ unit testing with automated unit test generation.
Supervisor: Assoc. Prof. Pham Ngoc Hung
 - * Directed the research group and facilitated weekly technical reviews with the joint development team at FPT Software.
 - * Provided training and technical consultation to group members.
 - * Researched and implemented an optimized pairwise test case generation and integrated it with the classification tree feature.
 - * Developed a method to stub functions from built-in or external dependencies.

PUBLICATIONS

Duy-Thanh Vu, **Duong Nguyen Vu Binh**, Oliver Y. Chén, Huong Ha & et al. *Bridging Kernel and Feature Spaces: A New Approach to Accurate and Interpretable Alzheimer's Disease Assessment*. In preparation.

Duong Nguyen Vu Binh, Duy-Thanh Vu, Duy-Cat Can, Hai Dac Nguyen, Binh T. Nguyen, Oliver Y. Chén & Huong Ha. *A Comparison of Machine Learning Methods for Alzheimer's Disease Classification in Vietnamese Patients*. International Symposium on Information and Communication Technology (SOICT 2025). Accepted.

Nguyen, M., **Duong, N. V. B.**, Duc, V. T., Phan, C., Phan, T., & Ha, H. (2024, July). *Unveiling Heterogeneity in Early Alzheimer's Disease: A Data-Driven Exploration of Mild Cognitive Impairment and Normal Cognition*. In International Conference on the Development of Biomedical Engineering in Vietnam (pp. 408-427). Cham: Springer Nature Switzerland.

PRESENTATIONS

Oral Presentation

- “Unveiling Heterogeneity in Early Alzheimer’s Disease: A Data-Driven Exploration of Mild Cognitive Impairment and Normal Cognition.” at 10th International Conference on the Development of Biomedical Engineering in Vietnam
- “A Comparison of Machine Learning Methods for Alzheimer’s Disease Classification in Vietnamese Patients” at International Symposium on Information and Communication Technology (SOICT 2025).

Poster Presentation

- “Machine Learning-Based Analysis of Brain Atrophy Patterns in Alzheimer’s Disease: A Comparative Study Between ADNI and Vietnamese Cohorts” at International Conference on Knowledge and Data Acquisition and Applications & Asia-Pacific Symposium on Affective Science and Alzheimer’s Disease (IC-KDA & ASAD 2025).

AWARDS AND HONORS

IC-KDA Asian Society Against Dementia Conference Travel Grant, Korea 2025
Vingroup Innovation Foundation Master’s Scholarship, Vietnam 2024 – 2025
Second prize, The Scientific Research Conference for Undergraduates at VNU-UET, Vietnam2023

TEACHING

Advanced Programming (INT2215) | Lecturer 2025 – Present
Object-Oriented Programming (INT2204) | Lecturer 2025 – Present
Object-Oriented Programming (INT2204) | Lab Instructor 2023 – Present
Introduction to Informatics (INT1004 & INT1376) | Lab Instructor 2023 – 2025

TECHNICAL SKILLS

MRI processing: CAT12, CONN
Programming: Python, JavaScript, C++, LaTeX
Languages: Vietnamese (native), English (IELTS 7.5 and TOELF IBT 106/120)

REFERENCES

Assoc. Prof. Pham Ngoc Hung

Vice Dean | Faculty of Information Technology, University of Engineering and Technology, Vietnam National University

Email: hungpn@vnu.edu.vn

Assoc. Prof. Huong Ha

Chair | Department of Tissue Engineering and Regenerative Medicine, School of Biomedical Engineering, International University, Vietnam National University Ho Chi Minh City

Email: htthuong@hcmiu.edu.vn